1. ABSTRACT

The project “Employee Performance Analysis” has been developed as part of the industrial training program on Data Analytics and Visualization with AI and Power BI. The objective of this project is to analyse employee performance, productivity, and attrition patterns within an organization using data analytics techniques and interactive data visualization.

The project utilizes HR analytics datasets to extract meaningful insights related to employee demographics, departmental distribution, workload, training impact, and workforce retention. It is used to transform raw data into an interactive dashboard that presents key performance indicators such as total employees, total projects handled, average income, average working hours, and attrition rate.

Various visualization techniques including bar charts, column charts, donut chart, line chart, area chart, map, matrix and slicers are used to enhance the analytical capabilities of the dashboard. Additionally, the dashboard incorporates a Dark Mode and Light Mode toggle feature, allowing users to switch themes for better visual comfort and improved user experience.

This dashboard helps organizations understand workforce trends, identify factors influencing employee performance and attrition, and support effective data-driven HR decision-making.

Keywords: Data Analytics, HR Analytics, Attrition Analysis, Data Visualization, Business Insights.

2. INTRODUCTION

2.1. Overview:

In modern organizations, analysing employee data is essential for understanding workforce performance, productivity, and employee retention patterns. With the growing importance of data-driven decision-making, businesses rely on analytics tools to transform raw HR data into meaningful insights.

This project analyses employee information such as department distribution, workload, salary patterns, training impact, and attrition trends. Through interactive visualizations, the dashboard enables users to explore workforce data effectively and identify key patterns that can support strategic HR decisions.

2.2 Key Features of the Dashboard:

The dashboard provides several analytical features to understand employee performance and organizational trends. Some of the key features include:

- **Employee Overview Analysis -** showing employee distribution across departments, gender, and locations.
- **Performance Analysis** - highlighting total projects handled, average working hours, role-level contributions, training impact and other.
- **Attrition Analysis** - identifying factors influencing employee turnover and attrition rate.
- **KPI Indicators -** displaying important metrics such as total employees, average income, average years at company, total projects, and attrition rate.
- **Department-wise Workforce Insights** - to analyse employee distribution across different departments.

2.3 Interactive Capabilities

The dashboard includes several interactive components that enhance user experience and allow deeper exploration of the data:

- **Dynamic Slicers and Filters -** for Age Group, Department, Job Role, Work-Life Balance, and Gender.
- **Interactive Visualizations** - such as bar charts, column charts, donut charts, line charts, area charts and maps to represent workforce insights.
- **Cross-filtering between visuals** - enabling users to analyse relationships between different variables.
- **Dark Mode and Light Mode toggle feature** - allowing users to switch between themes for better visual comfort and accessibility.

Overall, the Employee Performance Analysis provides an interactive view of workforce analytics, helping organizations understand employee performance trends and support data-driven HR decision-making.

3.OBJECTIVES

The following are the objectives of the project:

i. To determine the **total number of employees** within the organization and provide a general workforce overview.
ii. To analyse the **gender distribution of employees** to understand workforce diversity.
iii. To identify **department-wise employee distribution** and determine which department has the highest number of employees.
iv. To examine the **average years employees spend at the company and Avg income**, providing insights into workforce stability, salary trends and tenure.
v. To analyse **employee distribution across different locations** using geographic visualization.
vi. To evaluate the **total number of projects handled by employees**, **average weekly working hours** indicating overall workforce productivity.
vii. To analyse **key employee metrics** such as total projects handled, performance rating, absent days, monthly rate, and average weekly working hours vary across employees.
viii. To examine the **contribution of employees based on role level** in terms of projects handled.
ix. To analyse the **relationship between training sessions and performance ratings** to understand the impact of employee development programs.
x. To study the **relationship between absence days and performance ratings** to evaluate whether absenteeism influences employee performance.
xi. To determine the **total number of employees who left the organization** and calculate the overall attrition rate.
xii. To identify **department-wise attrition trends** to understand which departments experience the highest employee turnover.
xiii. To analyse **attrition patterns based on salary slabs** to determine whether compensation levels influence employee retention.
xiv. How does **overtime influence employee attrition** within the organization.
xv. To examine **attrition trends based on years at the company** to understand when employees are most likely to leave the organization.

4. IMPLEMENTATION

4.1 Problem Definition

Organizations collect large volumes of employee-related data, but analysing this data effectively to understand workforce performance and attrition patterns can be challenging. Traditional reports and spreadsheets often fail to provide clear insights into employee productivity, workload distribution, training impact, and turnover trends. Without proper visualization and analysis tools, it becomes difficult for HR teams to identify performance gaps, monitor employee engagement, and make informed decisions. Therefore, this project aims to develop an **Employee Performance Analysis** that transforms raw HR and employee performance data into meaningful visual insights, enabling organizations to better understand workforce trends and support data-driven decision-making.

4.2 Feasibility Study

A feasibility study was conducted to evaluate whether the project could be successfully implemented using the available resources and tools.

i. Technical Feasibility:

The project is technically feasible as it uses Microsoft Power BI, a powerful data visualization tool capable of handling structured datasets and creating interactive dashboards. The available HR analytics and employee performance datasets provide sufficient information for meaningful analysis and visualization.

ii. Operational Feasibility:

The dashboard is designed to be interactive and easy to use, allowing users to explore insights through charts, slicers, and filters. This enables HR professionals and managers to quickly analyse workforce trends without requiring advanced technical knowledge.

iii. Economic Feasibility:

The project requires minimal cost since it utilizes existing datasets and commonly used analytics tools. Power BI enables efficient data analysis and visualization without the need for expensive infrastructure or additional software resources.

4.3 Steps

The Project includes the following steps:

Step 1: Data Collection

The dataset used in this project was collected from **Kaggle**, which provides publicly available datasets for data analysis and visualization projects. [Kaggle - https://www.kaggle.com/]

• **HR Analytics Table:** Contains employee-related information such as EmpID, Age, Department, Job Role, Gender, Work Experience, Salary details, and Attrition etc.

EmpID	Age	AgeGroup	Attrition	BusinessTravel	DailyRate	Department	Location	DistanceFromHome	Education	EducationField	EmployeeNumber	EnvironmentSatisfaction	Gender	Hou
RM010	19	18-25	No	Travel_Rarely	1181	Research & Development	Sydney	3	1	Medical	201	2	Female	
RM013	19	18-25	Yes	Travel_Rarely	489	Human Resources	Chicago	2	2	Technical Degree	566	1	Male	
RM014	19	18-25	Yes	Travel_Rarely	419	Sales	London	21	3	Other	959	4	Male	
RM019	20	18-25	No	Travel_Rarely	959	Research & Development	Toronto	1	3	Life Sciences	657	4	Female	
RM022	20	18-25	Yes	Travel_Rarely	129	Research & Development	Sydney	4	3	Technical Degree	960	1	Male	
RM023	20	18-25	Yes	Travel_Rarely	1097	Research & Development	Toronto	11	3	Medical	1016	4	Female	
RM033	21	18-25	Yes	Travel_Rarely	156	Sales	Toronto	12	3	Life Sciences	494	3	Female	
RM035	21	18-25	Yes	Travel_Rarely	1427	Research & Development	Toronto	18	1	Other	923	4	Female	
RM042	22	18-25	No	Non-Travel	1123	Research & Development	Sydney	16	2	Medical	22	4	Male	
RM055	22	18-25	Yes	Travel_Rarely	1294	Research & Development	Singapore	8	1	Medical	1783	3	Female	
RM056	22	18-25	Yes	Travel_Rarely	391	Research & Development	New York	7	1	Life Sciences	1878	4	Male	
RM080	24	18-25	No	Travel_Rarely	823	Research & Development	New York	17	2	Other	643	4	Male	
RM083	24	18-25	No	Non-Travel	1092	Research & Development	Singapore	9	3	Life Sciences	812	3	Male	
RM087	24	18-25	Yes	Travel_Rarely	984	Research & Development	Toronto	17	2	Life Sciences	1219	4	Female	
RM090	24	18-25	No	Non-Travel	830	Sales	London	13	2	Life Sciences	1495	4	Female	
RM094	24	18-25	Yes	Travel_Rarely	240	Human Resources	London	22	1	Human Resources	1714	4	Male	
RM111	25	18-25	Yes	Travel_Rarely	867	Sales	Singapore	19	2	Marketing	952	3	Male	
RM137	26	26-35	Yes	Travel_Rarely	471	Research & Development	Singapore	24	3	Technical Degree	622	3	Male	
RM154	26	26-35	No	Travel_Rarely	474	Research & Development	Sydney	3	3	Life Sciences	1581	1	Female	
RM165	27	26-35	No	Travel_Rarely	1240	Research & Development	Toronto	2	4	Life Sciences	54	4	Female	
RM195	27	26-35	No	Travel_Rarely	1291	Sales	London	11	3	Medical	1364	3	Female	
RM209	27	26-35	Yes	Travel_Frequently	1337	Human Resources	Sydney	22	3	Human Resources	1944	1	Female	
RM230	28	26-35	No	Travel_Rarely	1144	Sales	Chicago	10	1	Medical	1056	4	Male	
RM234	28	26-35	Yes	Travel_Frequently	1009	Research & Development	Sydney	1	3	Medical	1111	1	Male	
RM248	28	26-35	No	Travel_Rarely	1423	Research & Development	Sydney	1	3	Life Sciences	1506	1	Male	
RM256	28	26-35	No	Travel_Rarely	1172	Sales	New York	3	3	Medical	1875	2	Female	
RM283	29	26-35	No	Travel_Rarely	1396	Sales	Sydney	10	3	Life Sciences	749	3	Male	

Table: HR_Analytics (1,480 rows)

Fig-4.1: HR_Analytics Dataset.csv

• **Employee Performance Table:** Includes performance-related attributes such as role level, performance rating, average working hours, number of projects handled, and absence days.

EmpID	role_level	avg_weekly_hours	projects_handled	performance_rating	absences_days
RM002	Senior	36	2	4	13
RM007	Senior	59	6	4	18
RM014	Senior	47	7	4	2
RM019	Senior	47	7	4	3
RM078	Senior	42	8	4	12
RM088	Senior	49	6	4	13
RM108	Senior	36	5	4	11
RM120	Senior	41	2	4	15
RM124	Senior	42	2	4	1
RM138	Senior	54	4	4	5
RM165	Senior	57	1	4	13
RM186	Senior	47	4	4	4
RM189	Senior	39	6	4	20
RM193	Senior	43	4	4	14
RM194	Senior	51	5	4	0
RM202	Senior	41	3	4	16
RM282	Senior	49	4	4	3
RM286	Senior	54	1	4	1
RM295	Senior	58	5	4	19
RM301	Senior	65	3	4	10
RM303	Senior	64	3	4	3
RM305	Senior	38	7	4	14
RM310	Senior	58	7	4	14
RM319	Senior	53	4	4	19
RM325	Senior	45	1	4	9
RM344	Senior	57	5	4	18
RM354	Senior	39	8	4	6

Table: employee_performance (1,480 rows)

Fig-4.2: Employee_performance Dataset.csv

Step 2: Data Cleaning

Data cleaning was performed using **Power Query Editor** in Power BI to ensure the dataset is accurate and suitable for analysis.

• Removed unnecessary columns such as Over18, as it contained only a single constant value ("Y") and did not add analytical value.

• Deleted columns like StockOptionLevel that were not relevant to the objectives of the dashboard.

• Checked and removed duplicate records using the *Remove Duplicates* option in Power Query to maintain data integrity.

• Verified and corrected data types (e.g., numeric, text, whole number) for columns like MonthlyIncome, Age, and Working Years to ensure proper calculations.

• Ensured consistency in categorical fields such as Department, Job Role, and Gender.

• Merged datasets using EmpID as the key column to establish proper relationships between tables

Step 3: Data Modelling

Data modelling was performed in Power BI to establish relationships between the datasets and ensure efficient analysis.

• Two tables were used: HR Analytics Table (Fact Table) and Employee Performance Table, connected using EmpID with a one-to-one relationship.

• The HR Analytics table contains core employee and attrition data, while the Employee Performance table includes metrics such as projects handled, working hours, and absence days.

• This relationship ensures accurate data filtering and seamless interaction across visuals in the dashboard.

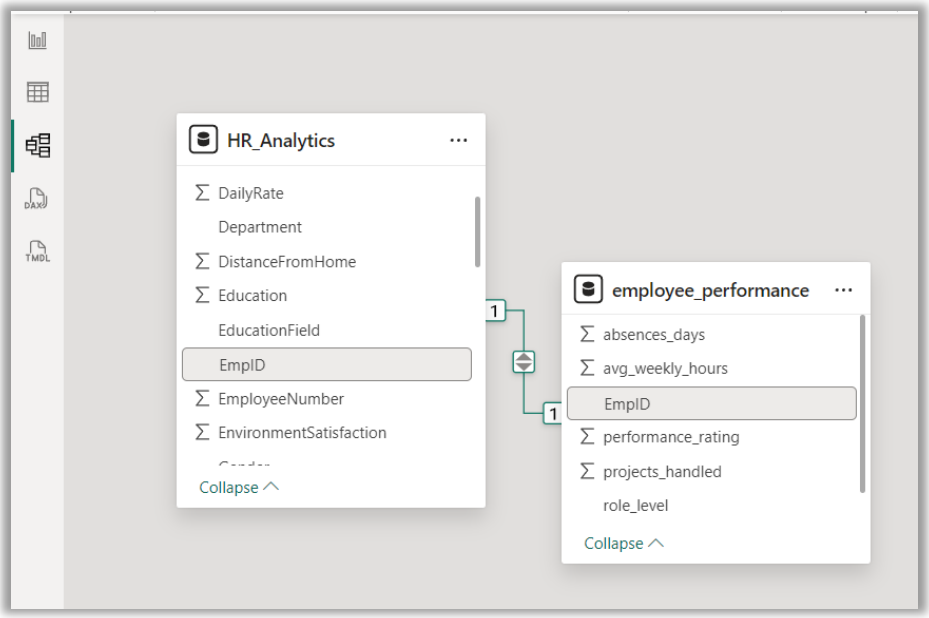

Fig-4.3: Data Relationship

Step 4: Data Analysis and Visualization

i. Determine the total number of employees within the organization:

• Created a measure using DAX:

```
Total Employees = DISTINCTCOUNT('HR_Analytics'[EmpID])
```

• Displayed this measure using a Card visual to show the total number of employees.

• This KPI provides a quick overview of the total workforce size in the organization.

Fig-4.4: Total Employees

ii. Analyse the Gender distribution of employees:

• Used a **Donut chart** to visualize the distribution of employees based on gender.

• Added **Gender** as legend and **Total Employees** as values to compare proportions, and change the formatting as needed.

• This analysis helps in understanding the **workforce diversity** within the organization.

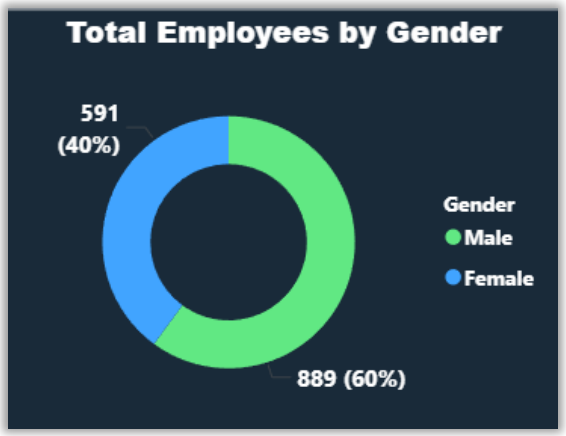

Fig-4.5: Total Employees by Gender

iii. Identify Department-wise employee distribution:

• Created a **Clustered Bar Chart** and added **Department** to the Y axis and **Total Employees** to X axis.

• This visualization helps in identifying which departments have the **highest and lowest number of employees**, giving a clear view of workforce distribution.

• Formatted the visual by adjusting **colours, data labels, and axis titles** to improve readability and maintain consistency with the dashboard theme.

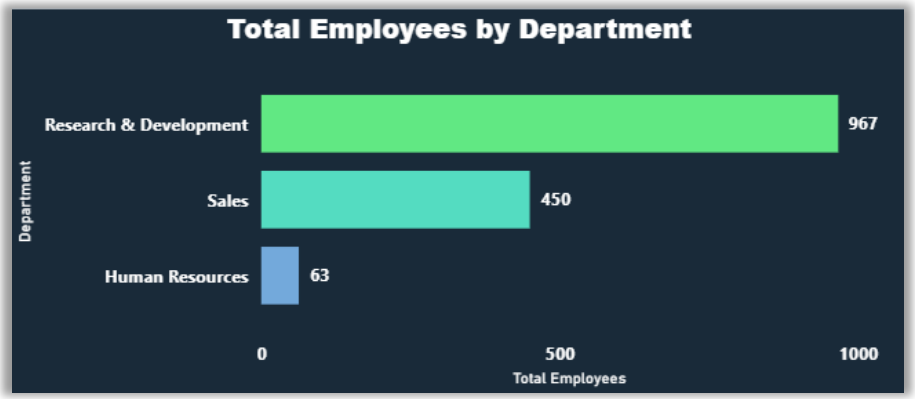

Fig-4.6: Total Employees by Department

iv. Determine average years employees spend at the company and Avg income:

• Used the existing columns **YearsAtCompany** and **MonthlyIncome** directly in visuals.

• Changed the **aggregation to Average** in the field settings.

• Displayed these values using **Card visuals** to show average employee tenure and income.

• These KPIs help in understanding **employee retention trends** and overall **salary distribution** in the organization.

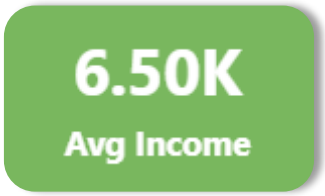

Fig-4.7: Average Income

Fig-4.8: Average years at company

v. Analyse employee distribution across different locations:

• Created a **Map visual** and added the **Location** column in location field along with **Total Emp in bubble size**.

• This visualization shows the **geographical distribution of employees** across different locations.

• Applied necessary **formatting** to improve clarity and overall presentation.

Fig-4.9: Total employees by location

vi. Evaluate the total number of projects handled by employees, average weekly working hours:

• Created a DAX measure:

```
Total Projects = SUM('employee_Performance'[projects_handled])
```

• Used **Average Weekly Hours** column and set aggregation to **Average**.

• Displayed both metrics using **Card visuals** for quick performance insights.

Fig-4.10: Total Project

Fig-4.11: Average weekly hours

vii. Examine the contribution of employees based on role level in terms of projects handled:

• Created a **Clustered Bar Chart** with **Job Role Level** on the Y axis and **Total Projects** on the X axis.

• This visualization helps to compare how employees at different levels contribute in terms of **projects handled**.

• Applied basic **formatting** to improve readability and maintain dashboard consistency.

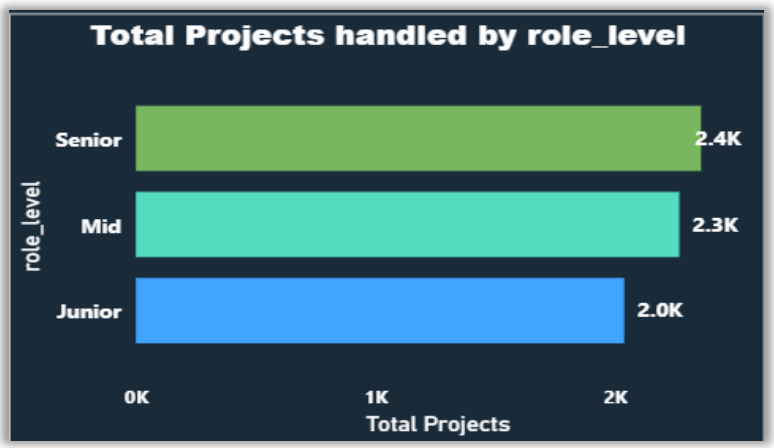

Fig-4.12: Total project handled by role level

viii. Analyse the relationship between training sessions and performance ratings:

• Created a **Clustered Column Chart** with **Training Sessions** on the Y axis and Avg **Performance Rating** on the X axis.

• This visualization helps to analyse the **relationship between training and employee performance.**

• Applied necessary **formatting** to enhance clarity and presentation.

Fig-4.13: Performance rating vs Training time

ix. Analyse the relationship between absence days and performance ratings:

• Created a **Line Chart** with **Absence Days** on the X axis and Avg **Performance Rating** as Y axis.

• This visualization helps to observe the **trend between absenteeism and performance levels.**

• Applied basic **formatting** to improve clarity and consistency of the chart.

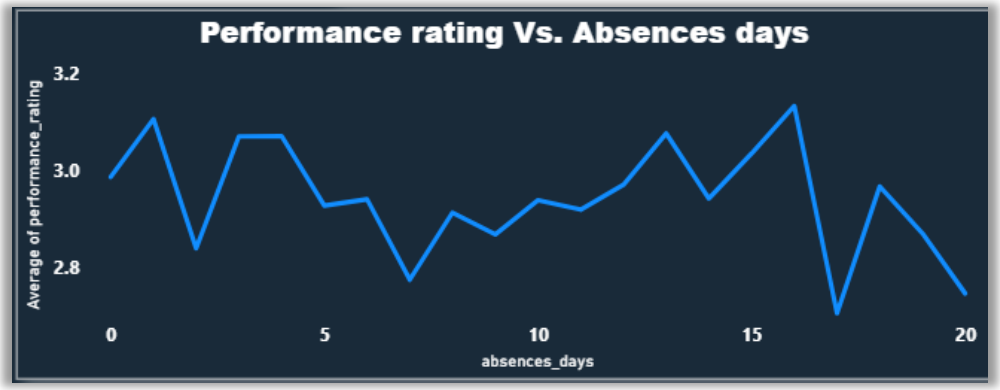

Fig-4.14: Performance rating vs Absence days

x. Analyse key employee metrics:

• Created a **Matrix visual** and added **Employee ID** in rows to display employee-wise data.

• Included fields such as **Total Projects, Performance Rating, Absent Days, Monthly Rate, and Average Weekly Hours** in values.

• This visualization helps to analyze how key performance metrics **vary across individual employees.**

EmpID	Total Projects	Performance_rating	Avg_weekly_hours	Absences_days	MonthlyRate
RM001	1	1	55	17	25233
RM002	2	4	36	13	9724
RM003	2	2	61	20	8059
RM004	4	1	36	17	13493
RM005	5	5	53	5	13556
RM006	2	5	38	1	19305
RM007	6	4	59	18	18420
RM008	5	5	37	15	8018
RM009	7	2	38	10	26820
Total	**6668**	**4358**	**73779**	**14453**	**21161722**

Fig-4.15: Employee metrics

xi. Attrition count and Attrition rate:

• Created DAX measures:

```
Attrition Count = CALCULATE([Total Employees],'HR_Analytics'[Attrition]="Yes")
```

```
Attrition Rate % = DIVIDE([Attrition Count],[Total Employees])
```

• Displayed these measures using **Card visuals** to highlight overall attrition

Fig-4.16: Attrition count

Fig-4.17: Attrition Rate

xii. Identify department-wise attrition trends:

• Created a **Stacked Column Chart** with **Department** on the X axis and **Attrition count on Y axis.**

• This visualization helps to identify **attrition patterns across different departments.**

• Applied necessary **formatting** to improve clarity and comparison.

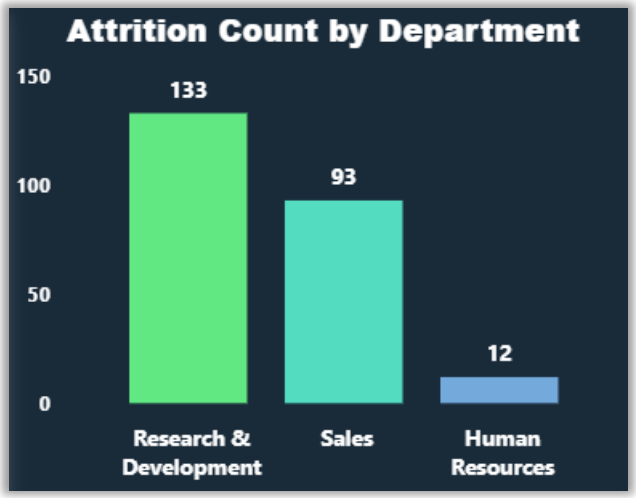

Fig-4.18: Attrition count by Department

xiii. Does overtime influence employee attrition:

• Created a **Stacked Column Chart** using **OverTime** as the X axis and **Attrition count on Y axis.**

• This visualization helps to analyze whether employees who work overtime are **more likely to leave the organization.**

• Applied necessary **formatting** to clearly compare attrition between overtime and non-overtime employees.

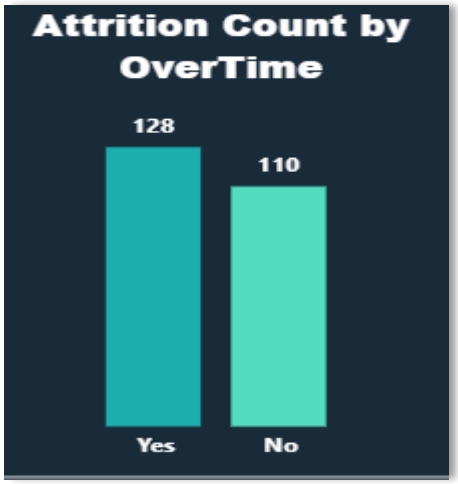

Fig-4.19: Attrition count by Overtime

xiv. Examine attrition trends based on years spend at the company:

• Created an **Area Chart** with **YearsAtCompany** on the X axis and **Attrition count on Y axis.**

• This visualization helps to examine how **attrition varies across different tenure levels.**

• Applied necessary **formatting** to improve clarity and highlight trends.

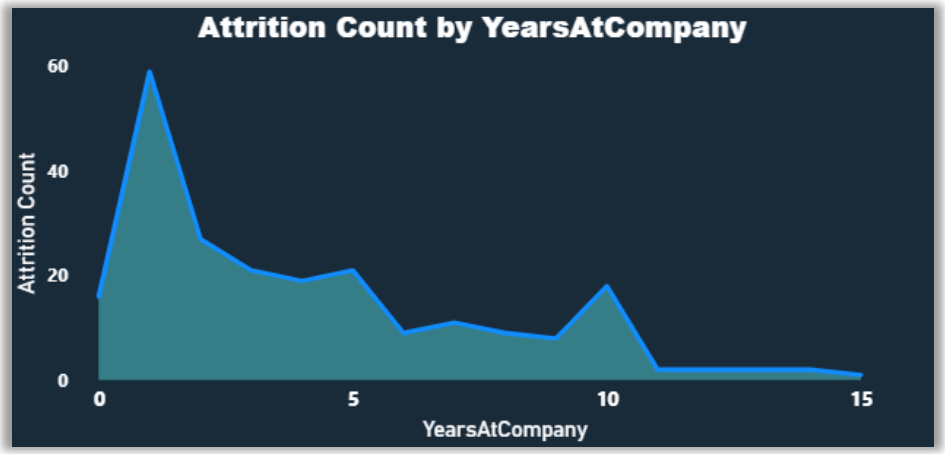

Fig-4.20: Attrition count by YearsAtCompany

xv. Analyse attrition patterns based on salary slabs:

• Created a Clustered Column Chart with **Salary Slab** on the X axis and **Attrition count on Y axis.**

• This visualization helps to analyze how attrition varies across different salary ranges.

• Applied necessary formatting to improve clarity and comparison.

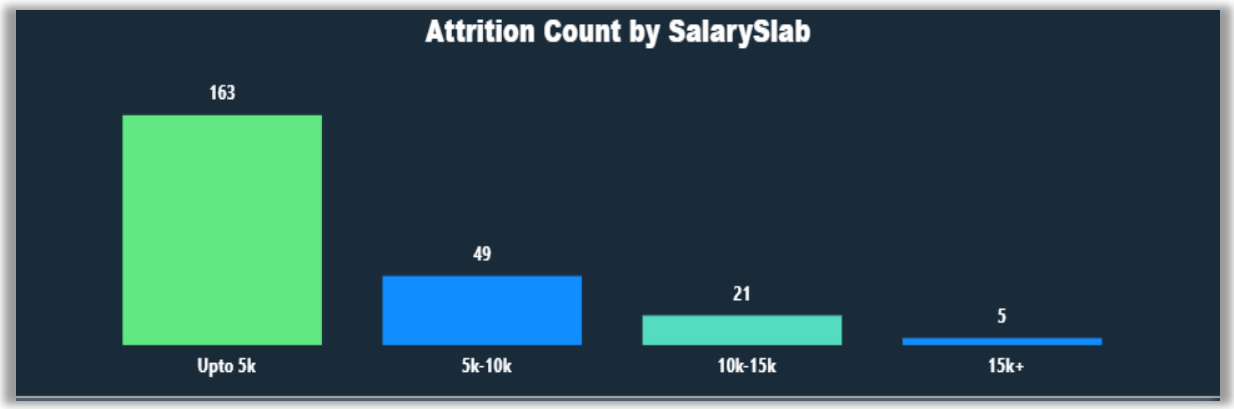

Fig-4.21: Attrition count by SalarySlab

• Interactive Features

• **Slicers:**

• Added slicers for Age Group, Department, Work-Life Balance, and Job Role.

• Enabled dynamic filtering to allow users to interactively explore the data.

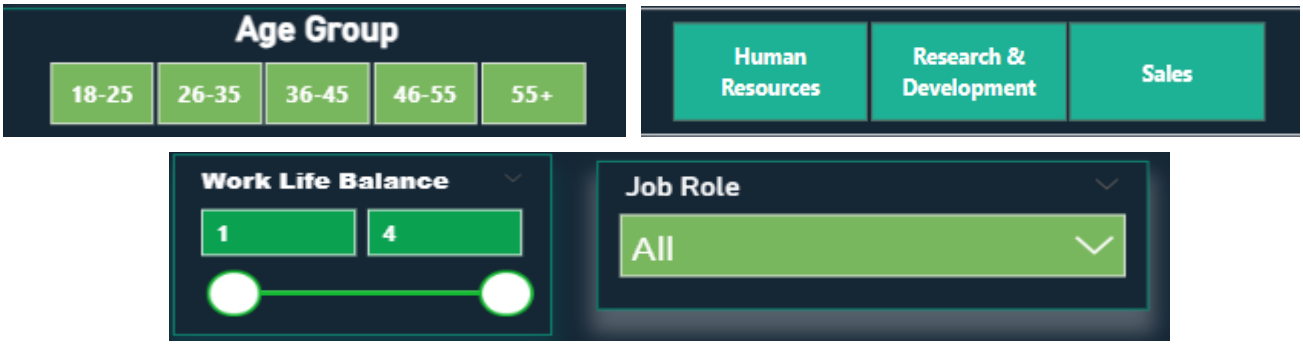

Fig-4.22: Slicers

- **Filters:**
    - Applied Gender as a filter across all report pages.
    - Ensured consistent and comparative analysis throughout the dashboard.

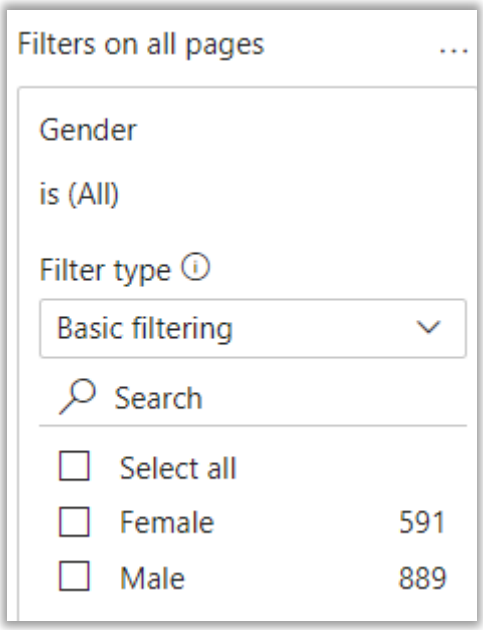

Fig-4.23: Filter on all pages

Step 5: Dashboard Creation

- **Theme Change (Dark Mode & Light Mode Toggle):**

  - Created a **Mode table** manually containing fields such as Mode (Dark/Light) along with corresponding **colour codes** for background, visual background, text, and shadow.

Mode	Backgroud	Visual_bg	Shadow	Text
Light	#E6F8EE	#F3FDF8	#076463	#112231
Dark	#112231	#192A39	#E6E6E6	#FFFFFF

Fig-4.24: Mode Table

- Added a **slicer** using the Mode field to allow users to switch between Light and Dark themes.

Fig-4.25: Slicer for Toggle

- Applied **conditional formatting (fx)** to all visuals by linking colour properties (background, text, titles, shadows) to the selected mode values from the table.

- For every colour choose field value as format style and choose respective column as 'what field should we base this on', choose summarisation value First / Last, click OK.
- Inserted a **shape covering the canvas** and applied conditional formatting to its fill colour to dynamically change the dashboard background as canvas background doesn't have any conditional formatting option by default.

- Sent the shape to the back layer so that all visuals remain visible while the background updates based on the selected mode.

- **Cover Page Creation:**
  - Created a new blank page in Power BI and set it as the cover page for the dashboard.
  - Inserted a text box to add the title *"Employee Performance Analysis"* and formatted it with appropriate font size and style.
  - Added a short description below the title using another text box to explain the purpose of the dashboard.
  - Inserted navigation buttons and enabled the *Action* feature to navigate between different report pages.
  - Add the dark mode & light mode toggle button with icons.
  - Applied formatting (colours, shapes, alignment, and spacing) to create a clean, consistent, and professional layout.

Fig-4.26: Cover page in dark mode

Fig-4.27: Cover page in light mode

- **Employee Overview Page Creation:**

- Created a **new report page** and added key visuals to analyse overall employee distribution and demographics.
- Added a **Gender Distribution chart**, **Department-wise Employee Count Chart**, and a **Map visual** to show employee distribution across locations.
- Inserted **KPI cards** to display Total Employees, Average Income, and Average Years at Company for quick insights.
- Added an **Age Group slicer** to enable dynamic filtering of employee data.
- Included a **Dark/Light mode slicer** to allow users to switch between themes interactively.
- Insert a right arrow button for the destination page performance overview.

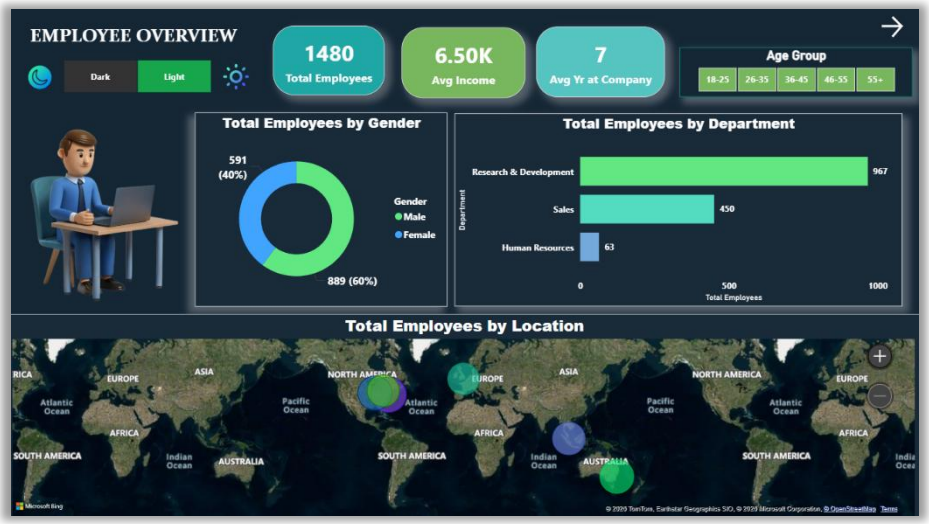

Fig-4.28: Employee overview page in dark mode

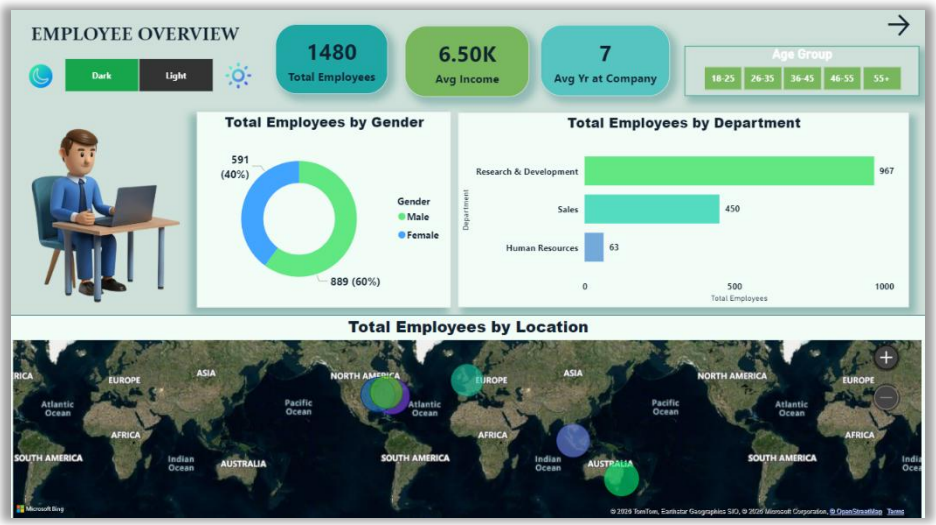

Fig-4.29: Employee overview page in light mode

- **Performance Overview Page Creation:**

- Created a new report page to analyse employee performance metrics and productivity.
- Added visuals such as Total Projects (Card), Average Weekly Hours (Card), and a Matrix to display employee-wise performance details.
- Included charts like Job Role Level vs Projects handled Chart, Training vs Performance rating Chart, and Absence vs Performance Chart for deeper analysis.
- Added relevant slicers and filters to enable interactive analysis across performance-related metrics.
- Included a **Dark/Light mode slicer** to allow users to switch between themes interactively.
- Insert a right arrow button for the destination page attrition overview.

Fig-4.30: Performance overview page in dark mode

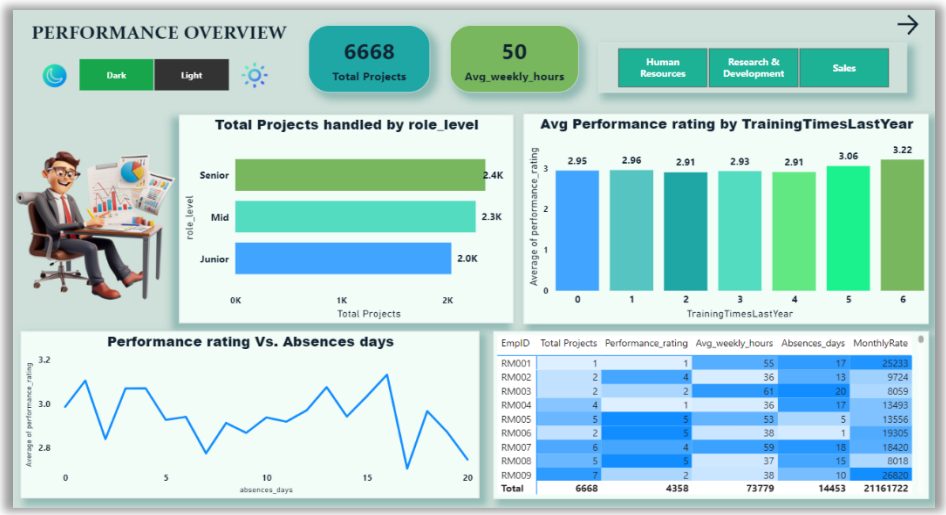

Fig-4.31: Performance overview page in light mode

- **Attrition Overview Page Creation:**
  - Created a **new report page** to analyse employee attrition trends and patterns.
  - Added **KPI cards** for Attrition Count and Attrition Rate to provide a quick overview of employee turnover.
  - Included visuals such as **Department-wise Attrition Chart, Overtime vs Attrition Chart, Attrition by Years at Company Chart**, and **Attrition by Salary Slab Chart.**
  - Added relevant **filters and slicers** to allow interactive analysis of attrition data.
  - Included a **Dark/Light mode slicer** to allow users to switch between themes interactively.

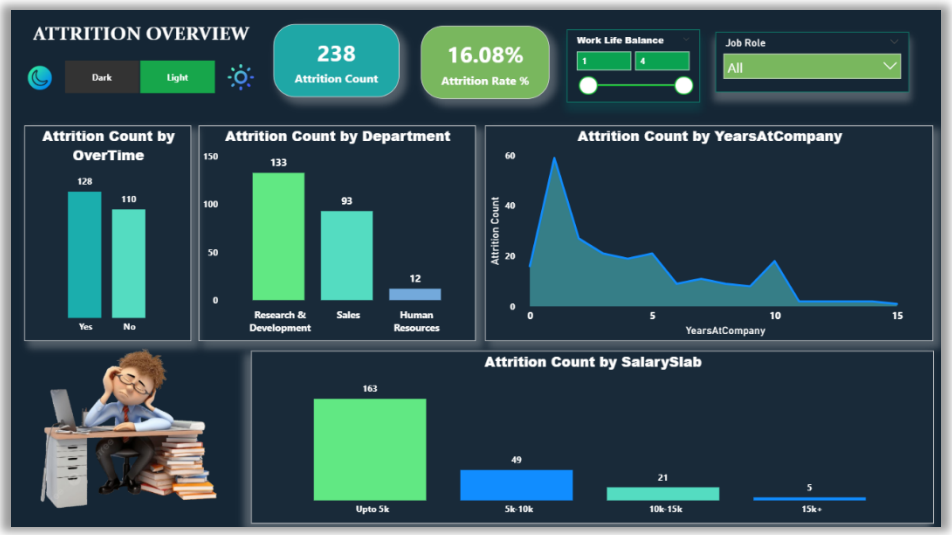

Fig-4.32: Attrition overview page in dark mode

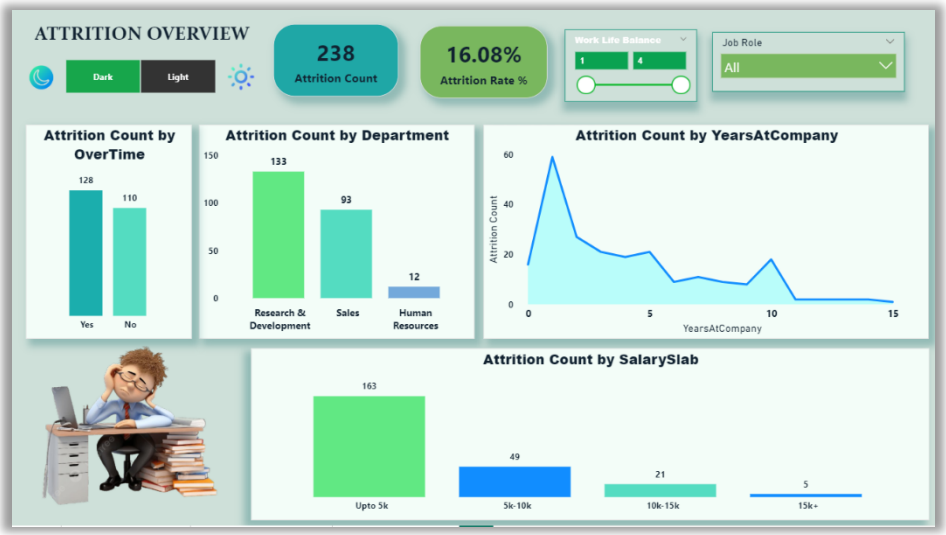

Fig-4.33: Attrition overview page in light mode

4.4 Testing and Validation

- **Data Accuracy:**

  The values displayed in various visuals such as cards, charts, and matrix tables were carefully cross-checked with the original dataset to ensure accuracy using Microsoft Power BI. Aggregations like sum, average, and count were verified to confirm correct calculations and to avoid any inconsistencies in the analysis.

- **Functionality Testing:**

  All interactive elements including slicers, filters, and navigation buttons were tested to ensure they function correctly. It was verified that selecting different options dynamically updates the visuals across the report pages without any errors or delays.

- **Usability Testing:**

  The overall dashboard layout, alignment, spacing, and readability were reviewed to ensure a smooth user experience. The design was checked to confirm that users can easily understand the visuals and navigate through different sections without confusion.

- **Consistency Testing:**

  The dashboard was checked to ensure consistency in formatting, colour themes, and data representation across all pages. It was verified that similar metrics display uniform values in different visuals and that the Light/Dark mode applies correctly throughout the report without any mismatch.

5. RESULT AND DISCUSSION

This section presents the **key findings** obtained from the analysis of the Employee Performance Analysis across different report pages.

5.1 Employee Overview Page Findings:

- The organization consists of a total of 1480 employees with a gender distribution of 60% male and 40% female.
- The Research & Development department has the highest number of employees, indicating its major role in the organization.
- The average tenure of employees is approximately 7 years, reflecting a relatively stable workforce.
- Employees are distributed across multiple locations, showing a geographically diverse presence.
- The average monthly income of employees is approximately 6.50K, indicating the overall salary level within the organization.

5.2 Performance Overview Page Findings:

- Employees have handled a total of 6668 projects, reflecting overall productivity levels.
- The average weekly working hours is approximately 50 hours, indicating the general workload of employees.
- Senior employees contribute more significantly to project handling compared to junior employees.
- A slight positive relationship is observed between training sessions and performance ratings.
- No strong or consistent relationship is found between absence days and performance ratings, indicating absenteeism alone doesn’t effect performance rating.

5.3 Attrition Overview Page Findings:

•A total of 238 employees have left the organization, resulting in an attrition rate of 16.08%.

• The highest attrition is observed in the Research & Development department.

• Employees in lower salary groups show higher attrition compared to higher salary groups.

• Most attrition occurs during the early years of employment.

• Employees working overtime are more likely to leave, indicating that overtime has a noticeable impact on attrition.

The analysis of the **Employee Overview** suggests that the company should focus on improving workforce balance and strategic planning. Efforts can be made to promote gender diversity through inclusive hiring practices. Since the Research & Development department has the highest number of employees, proper resource allocation and workload distribution should be ensured. Additionally, maintaining employee satisfaction through competitive compensation and engagement initiatives can help sustain the stable workforce.

The **Performance Overview** indicates that employee performance can be enhanced through structured training and development programs. The company should invest in continuous learning opportunities to further improve performance levels. Since senior employees contribute more significantly, mentorship programs can be introduced where experienced employees guide junior staff. Managing workload by monitoring average working hours can also help in maintaining productivity without causing burnout.

The **Attrition Overview** highlights the need for strong retention strategies. The company should focus on improving employee satisfaction, especially in lower salary groups, by reviewing compensation and benefits. High attrition in the early years suggests the need for better onboarding and engagement programs. Additionally, reducing excessive overtime and ensuring a healthy work-life balance can help minimize employee turnover and improve overall retention.

6. FUTURE SCOPE

The Employee Performance Analysis can be further improved by incorporating advanced analytics capabilities and expanding the available data sources.

• **Real-Time Data Integration:** Future versions of the dashboard can connect to live HR databases to monitor employee performance and workforce metrics in real time.

• **Predictive Analytics for Attrition:** Machine learning models can be integrated to predict employee attrition and identify employees who may leave the organization in the future.

• **Integration of Additional HR Datasets:** Adding datasets such as employee satisfaction surveys, recruitment data, and training effectiveness data can provide deeper workforce insights.

• **Cloud-Based Deployment:** The dashboard can be deployed on cloud platforms like Power BI Service to enable secure access, sharing, and collaboration across departments.

• **Mobile Dashboard Accessibility:** The dashboard can be optimized for mobile devices so that managers and decision-makers can monitor workforce insights anytime and anywhere.

• **Automation:** Use Python or SQL for automated data cleaning and preprocessing**.**

These enhancements can make the dashboard more powerful, scalable, and valuable for long-term workforce analytics and organizational planning.

7. APPLICATION AND CONCLUSION

7.1 Human Resource Management

- Helps HR teams monitor employee performance and workforce distribution.
- Supports better decision-making in recruitment and resource allocation.

7.2 Employee Performance Improvement

- Identifies factors affecting performance such as training and experience.
- Assists in designing effective training and development programs.

7.3 Attrition Management

- Helps in identifying key reasons behind employee turnover.
- Supports the development of strategies to improve employee retention.

7.4 Workforce Planning

- Enables analysis of employee distribution across departments and locations.
- Assists in optimizing workload and improving operational efficiency.

7.5 Strategic Decision Making

- Provides insights for management to take data-driven decisions.
- Helps in improving overall organizational performance and growth.

7.6 Work-Life Balance Monitoring

- Helps in analysing working hours and overtime patterns.
- Supports initiatives to reduce employee burnout and improve satisfaction.

7.7 Training Effectiveness Analysis

- Evaluates the impact of training programs on employee performance.
- Assists in improving the quality and focus of training sessions.

7.8 Organizational Performance Tracking

- Provides a consolidated view of key performance indicators.
- Helps in tracking progress and identifying areas for improvement.

The Employee Performance Analysis successfully provides a comprehensive analysis of employee distribution, performance, and attrition within the organization. Through effective use of data visualization and analytical techniques, the dashboard highlights key patterns such as the impact of experience on performance, the role of training in improving productivity, and the major factors influencing employee attrition. The insights derived from this analysis can support data-driven decision-making and help organizations improve workforce management, enhance employee performance, and develop effective retention strategies. Overall, the project demonstrates the practical application of data analytics and visualization tools in solving real-world business problems.

In conclusion, this project demonstrates how data analytics and visualization can transform raw data into meaningful and actionable insights. By leveraging tools like Power BI, complex employee data is simplified into interactive dashboards that support quick and effective decision-making. The project highlights the importance of data-driven strategies in improving employee performance, managing attrition, and optimizing workforce planning. It also reflects the practical application of analytical concepts learned during training, including data cleaning, modelling, and visualization. Overall, this work emphasizes the growing role of analytics in modern organizations and its ability to drive efficiency, productivity, and long-term growth.

8. REFERENCES

The following books and articles were referred to for understanding concepts related to data analytics, data visualization, and Power BI.

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